



SUPPLY CHAIN RESILIENCE TRACK

Supply Chain Disruption Assistant & Fleet Utilization Optimizer

Turning chaotic disruptions into coordinated, cost-saving decisions — in real time.

TEAM-FUTURE

The Problem

Disruptions cascade faster than teams can track them manually



Cascading disruptions

Weather events, port strikes, and geopolitical crises ripple across hundreds of active shipments at once — far beyond what dispatchers can track by hand.



Idle & overloaded fleets

While some routes sit gridlocked, trucks, containers, and vessels elsewhere sit idle — capacity goes to waste in both directions.



Cold chain blind spots

Vaccines and perishables are especially fragile. A single temperature excursion on any leg of the journey can spoil cargo worth \$500K or more.



DISCOVERED TOO LATE

\$500K+

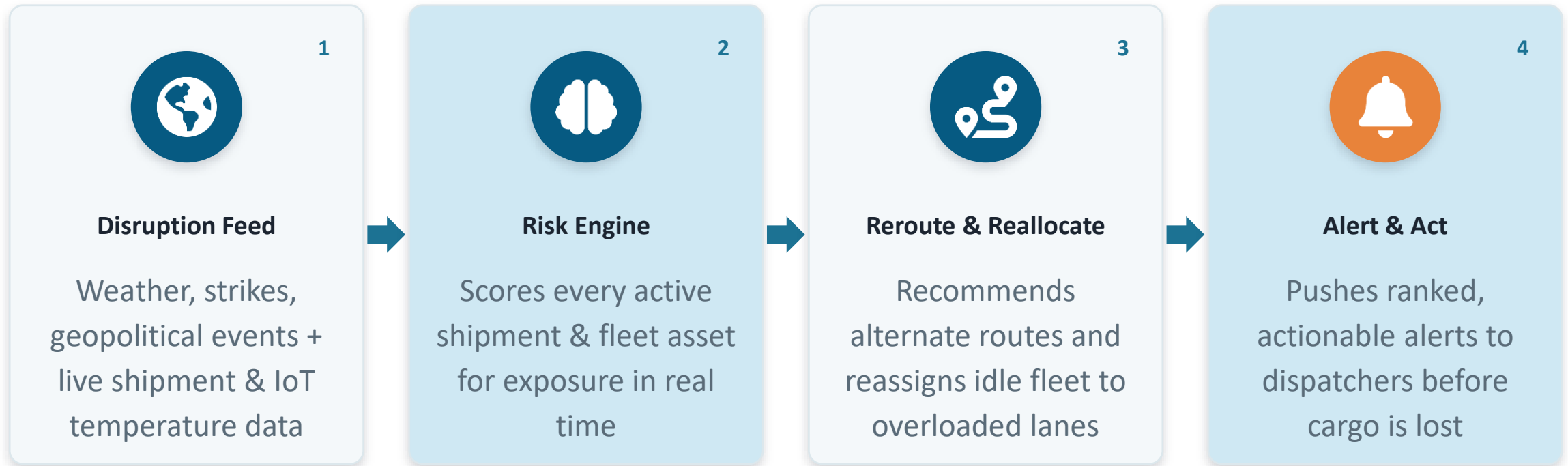
in cold-chain cargo value at risk per breach — and today, breaches are typically found only at delivery.

WHO FEELS IT

Logistics operators, fleet dispatchers, and pharma / cold-chain shippers who need to act within minutes, not after delivery.

Our Solution

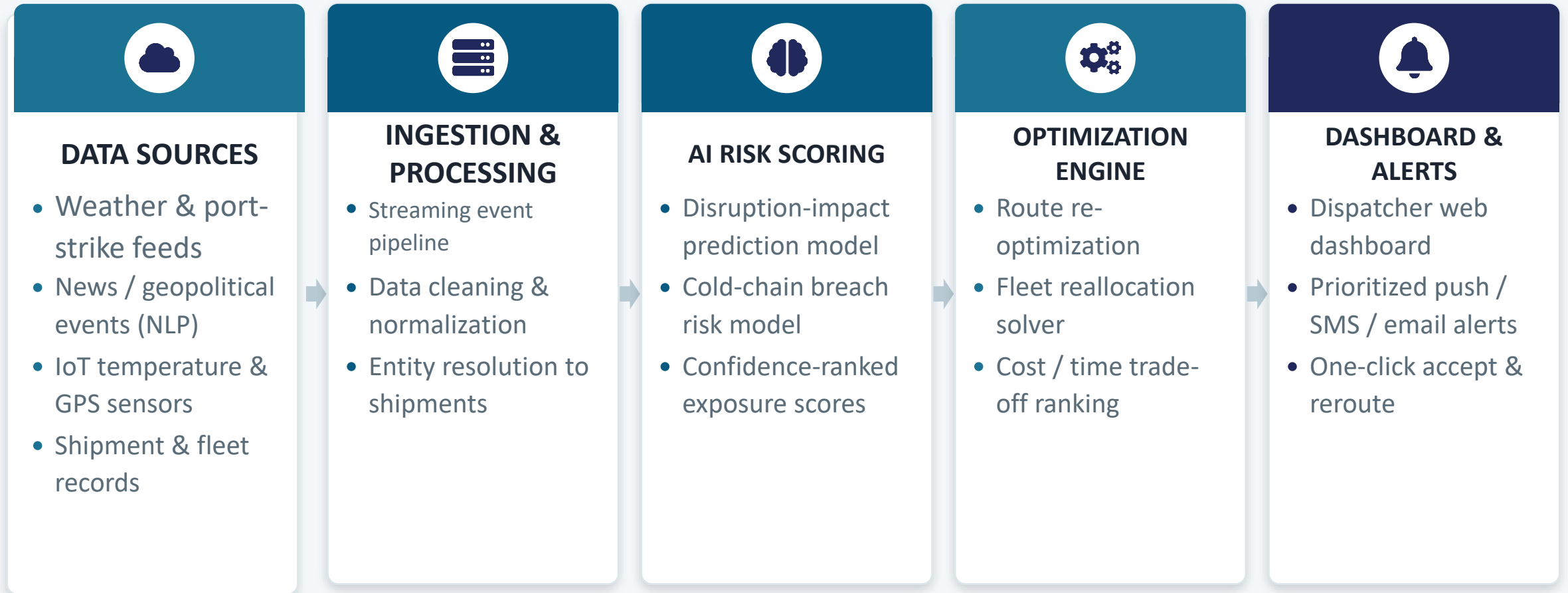
One pipeline: sense disruption → score risk → recommend action → alert the right people



Result: disruptions are caught upstream, fleet capacity is rebalanced automatically, and cold-chain cargo is protected before spoilage occurs.

Architecture

A layered pipeline from raw signals to dispatcher action



Demo: Cold-Chain Breach Prevention

Our strongest feature — catching a temperature excursion before it becomes a \$500K loss

Fleet & Shipment Command Center

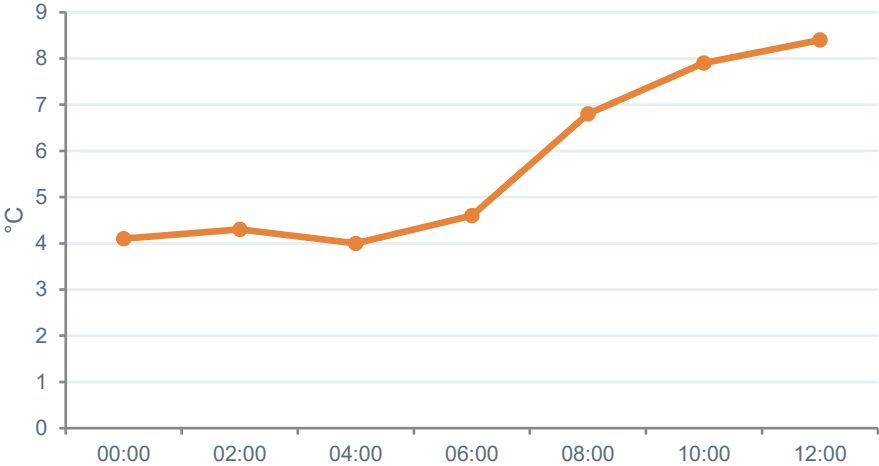


TEMPERATURE EXCURSION DETECTED — Shipment #CX-4471

Vaccine cargo, Leg 2 of 3 — cargo value \$612,000 — predicted breach in 42 min

Suggest Reroute

Cold-Chain Temperature Log



Recommended Actions (ranked)



Reroute via refrigerated Hub-7

Adds 38 min, saves full cargo value

BEST



Reassign nearest reefer truck

Idle unit 4.2 mi away, ETA 11 min

FAST



Notify receiving pharmacy now

Pre-empt delivery window change

ALERT

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5

IBM Technologies

[Edit: replace with the exact IBM products / services your build actually used]



watsonx.ai

Trains and serves the disruption-impact and cold-chain breach risk-scoring models.



Watson Discovery / NLP

Extracts disruption events (strikes, weather, unrest) from live news and reports.



IBM Db2 / Cloudant

Stores shipment, fleet, and sensor telemetry with low-latency reads for scoring.



IBM Cloud Functions

Serverless orchestration connecting ingestion, scoring, and alerting stages.



IBM Maximo

Tracks fleet asset status and utilization feeding the reallocation engine.



IBM Cloud Security

Identity, access control, and encryption across the data and API layers.

Results & Impact

[Edit: replace illustrative figures below with your actual pilot / test results]



80%

Faster disruption detection
vs. manual monitoring
baseline



92%

Breach prediction accuracy
on held-out cold-chain test
set



35%

Less idle fleet time
through automated
reallocation



\$2.3M

Cargo value protected
in simulated pilot scenarios

WHAT SUCCESS LOOKS LIKE

- No cold-chain shipment reaches its destination without an early temperature-risk warning.
- Idle fleet capacity is matched to overloaded lanes within minutes of a disruption event.
- Dispatchers get one ranked action, not a flood of raw alerts, cutting decision time to seconds.

The Team

[Edit: replace placeholder names, roles, and contributions]



Ansh Patel

ML & Risk Modeling

Built the disruption-impact and cold-chain breach prediction models using watsonx.ai.



Jeet patel

Frontend & Dashboard

Designed and built the dispatcher command center and alert workflow UI.



Kshitij Pandya

Data Engineering

Built the ingestion pipeline connecting weather, news, IoT, and shipment data.



Ansh Patwl

Cloud & DevOps

Set up IBM Cloud infrastructure, serverless orchestration, and deployment.